

Section 4.3

Let (X, d) be a Metric space, and let $f: X \rightarrow \mathbb{R}$ be a Continuous map.

Prove the zero set of f

$$Z(f) := \{x \in X \mid f(x) = 0\}$$

is closed.

Proof. Let $y \in \overline{Z(f)}$ be given. That is, for any $r > 0$,

$$B_r(y) \cap Z(f) \neq \emptyset.$$

Given $\varepsilon > 0$, there exists $\delta > 0$ s.t.

$$d(x, y) < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

For fixed $\delta > 0$, choose

$$x \in B_\delta(y) \cap Z(f)$$

Then $d(x, y) < \delta$ and $f(x) = 0$. Therefore

$$|f(y) - f(x)| = |f(y)| < \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, $f(y) = 0$. That is, $y \in Z(f)$.

Section 4.2

Let X, Y be Metric spaces. If $f: X \rightarrow Y$ is continuous, then for any $E \subset X$,

$$f[\overline{E}] \subseteq \overline{f[E]}.$$

Proof. Let $f(x) \in f[\overline{E}]$ be given. Since f is continuous, given $\varepsilon > 0$, there exists $\delta > 0$ s.t.

$$t \in B_\delta(x) \Rightarrow f(t) \in B_\varepsilon(f(x))$$

Meanwhile, since $x \in \overline{E}$, $B_\delta(x) \cap E \neq \emptyset$

Choose $u \in B_\delta(x) \cap E$. Then, $f(u) \in B_\varepsilon(f(x))$ and $f(u) \in f[E]$. That is

$$f(u) \in B_\varepsilon(f(x)) \cap f[E] \neq \emptyset.$$

This implies $f(x) \in \overline{f[E]}$.

Section 4.4

Let (X, d_X) and (Y, d_Y) be a Metric space, and $f, g: X \rightarrow Y$ be a continuous map.

If $E \subseteq X$ is a dense in X , then $f[E]$ is dense in $f[X]$.

Moreover, If $f|_E = g|_E$, then $f = g$.

Proof. Since f is continuous, $f[X] = f[\overline{E}] \subseteq \overline{f[E]}$ is clear.

Let $x \in X = \overline{E}$ be given. If $f(x) = g(x)$, there is nothing to prove.

Suppose $f(x) \neq g(x)$. Take $\varepsilon = d(f(x), g(x)) > 0$. Since f and g are continuous, there exist $\delta > 0$ s.t.

$$t \in B_\delta(x) \Rightarrow d(f(x), f(t)), d(g(x), g(t)) < \frac{\varepsilon}{2}.$$

Since E is dense in X , there is $t \in B_\delta(x) \cap E$. Now,

$$d(f(x), g(x)) \leq d(f(x), f(t)) + \underbrace{d(f(t), g(t))}_{=0} + d(g(t), g(x)) < \varepsilon = d(f(x), g(x))$$

It is Contradiction.